Arabidopsis Fruit Production Response to Nutrients and Damage

STAT 642 — Spring 2026 Team 11

Outline

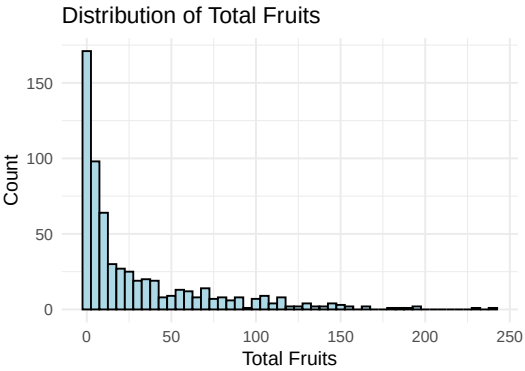
1. Introduction, Data, and Research Questions
2. Model and Assumptions
3. Model Reduction, Diagnostics, and Results
4. Conclusions

1. Introduction & Data

Source: `lme4::Arabidopsis` — a DoE experiment on *Arabidopsis thaliana* by the Center for Genomics and Systems Biology at NYU, published in Banta, Stevens, & Pigliucci (2010), *Oikos* 119(2): 359–369.

Banta et al. investigated how **nutrients** and **apical meristem damage (AMD)** interact to affect fruit production in plants sampled from three European regions, as a test of the "limiting resources" framework.

- **Sample size:** $n = 625$ plants
- **Response:** Total Fruits (zero-heavy count)
- **Unit of observation:** individual plant



2. Research Questions

Why it matters

- Plant response to AMD (herbivory / environmental stress) reflects growth, survival, adaptation
- Prior literature is split: higher nutrients $\rightarrow$ higher *or* lower tolerance
- This dataset enables a controlled test of that trade-off

Research Questions

1. How do the reproductive responses of *Arabidopsis thaliana* to AMD vary across regions, nutrient levels, and transplant sources?
2. Are there significant interactions between region, nutrient levels, and transplant sources in determining fruit production?
3. How much of the variability in fruit production can be attributed to differences between populations and maternal seed families within regions?

3. Variables & Experimental Design

Variable	Role	Levels / Description
Total Fruits	Response	count per plant
Region	Fixed	NL, SP, SW
Nutrients	Fixed	1 prill (low), 8 prills (high)
AMD	Fixed	clipped, unclipped
Status	Fixed	Normal, Petri.Plate, Transplant
Rack	Fixed (additive)	1, 2
Population	Random	9 levels, nested in Region
Genotype	Random	24 seed families, nested in Population

Design: 2×2 factorial treatment (Nutrients $\times$ AMD) crossed with a 3-level sampling hierarchy (Region $\rightarrow$ Population $\rightarrow$ Seed family). Status and Rack are fixed nuisance factors (Rack additive; Status interactions considered). $\approx$ 6.5 plants per seed-family $\times$ treatment cell.

4. Model

Fit via `lme4::lmer` with Satterthwaite F-tests (`lmerTest`). Response: $\log(\text{Total Fruits} + 1)$.

$$
\begin{aligned}
Y_{ijklmno} = \mu &+ \alpha_i + \beta_j + \gamma_k + \delta_l + \phi_m \\
&+ (\alpha\beta)_{ij} + (\alpha\phi)_{im} + (\beta\phi)_{jm} + (\alpha\beta\phi)_{ijm} \\
&+ (\alpha\gamma)_{ik} + (\beta\gamma)_{jk} + (\phi\gamma)_{mk} \\
&+ P_{n(i)} + F_{o(ni)} + \varepsilon_{ijklmno}
\end{aligned}
$$

Fixed effects

- μ: intercept; α_i: Region; β_j: Nutrients
- ϕ_m: AMD; γ_k: Status; δ_l: Rack
- Treatment interactions: $(\alpha\beta), (\alpha\phi), (\beta\phi), (\alpha\beta\phi)$
- Nuisance interactions: $(\alpha\gamma), (\beta\gamma), (\phi\gamma)$

Random effects

- $P_{n(i)} \sim N(0, \sigma_P^2)$: Population in Region
- $F_{o(ni)} \sim N(0, \sigma_F^2)$: Seed family in Population
- $\varepsilon_{ijklmno} \sim N(0, \sigma^2)$: residual

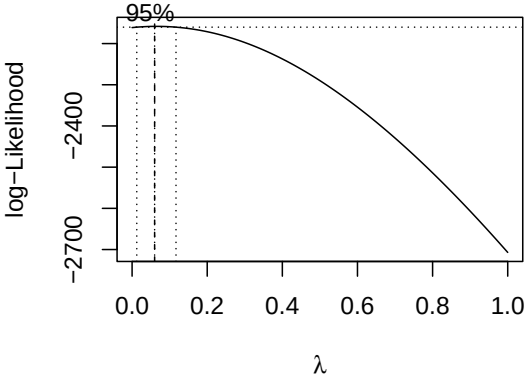
5. Assumptions & Response Transformation

LMM assumptions

1. **Linearity** (in predictors)
2. **Independence** (conditional on random effects)
3. **Normality** of random effects and residuals
4. **Equal variance** of residuals
5. **Correct specification** (fixed + random)
6. **Random sampling within strata** (populations, families)
7. **Exogeneity**: random effects uncorrelated with fixed effects

Response transformation

- Box-Cox $\lambda \approx 0.06 \approx 0 \Rightarrow \log$
- Use $\log(\text{Total Fruits} + 1)$ to handle zeros

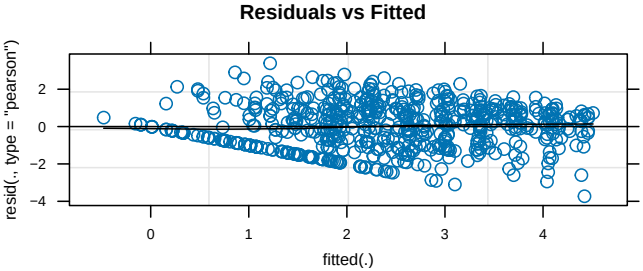


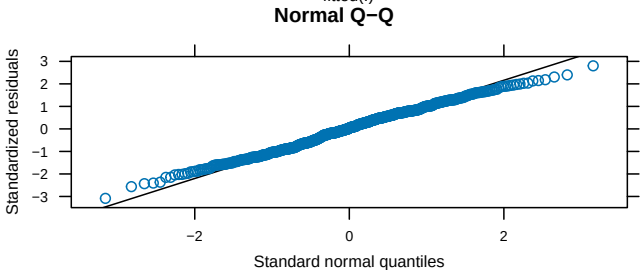
6. Model Reduction (LRT)

Backward elimination via LRT ($\alpha = 0.05$) on ML-fit models; drop nuisance interactions first, then highest-order treatment interaction, then remaining treatment two-ways. **Region × Nutrients** and **Region × Status** are retained. AIC all-subsets search (72 models, hierarchy enforced) prefers adding Nutrients × AMD — not LRT-significant, so dropped for interpretability.

Step	Interaction dropped	Chisq	Df	p
1	Nutrients × Status	0.89	2	0.640
2	AMD × Status	0.96	2	0.618
3	Region × Nutrients × AMD	2.69	2	0.261
4	Region × AMD	0.42	2	0.812
5	Nutrients × AMD	2.44	1	0.118

7. Diagnostics





Levene test on residuals

Factor	p_value
Region	0.0059
Nutrients	0.7100
Status	0.0245
AMD	0.3870

Summary

- Normality: Q–Q approximately linear; Shapiro p $\approx$ 0.028 reflects large n + zero-inflation
- Variance: mild heteroscedasticity across Region and Status; variance ratios < 1.5
- NB GLMM (slide 12) confirms robustness

8. Type III F-tests — Reduced Model

Table 4: Type III F-tests for log(Total Fruits + 1)

Effect	F	df1	df2	p-value
reg	2.781	2	6.41	0.135
nutrient	148.973	1	600.74	<0.001
amd	11.102	1	597.19	<0.001
status	2.661	2	606.10	0.071
rack	57.176	1	598.90	<0.001
reg:nutrient	4.768	2	599.90	0.009
reg:status	2.514	4	605.95	0.041

Significant effects

- Main: **Nutrients, AMD, Rack, Status**
- Interactions: **Region × Nutrients**, **Region × Status**

9. Variance Components (RQ3)

We partition the random-effect variance from the reduced LMM and test each component via LRT.

Table 5: Variance components (reduced LMM)

Component	Variance	% of total
gen	0.024	1.3
popu	0.289	16.3
Residual	1.463	82.4

Table 6: LRT for random effects

Term	LRT	Df	p-value
	NA	NA	NA
(1 | popu)	12.97	1	<0.001
(1 | gen)	1.48	1	0.224

Approximately **16.3%** of the variance in fruit production is attributable to differences between populations. The LRT for maternal seed families is not significant, so that level does not explain a meaningful portion of the variability.

10. Post-hoc Analysis (EMMs)

Pairwise contrasts on the $\log(\text{Total Fruits} + 1)$ scale.

Nutrients within Region

Reg	Contrast	Est.	SE	p
NL	nutrient1 - nutrient8	-1.562	0.232	0
SP	nutrient1 - nutrient8	-1.458	0.142	0
SW	nutrient1 - nutrient8	-0.859	0.165	0

AMD (main effect)

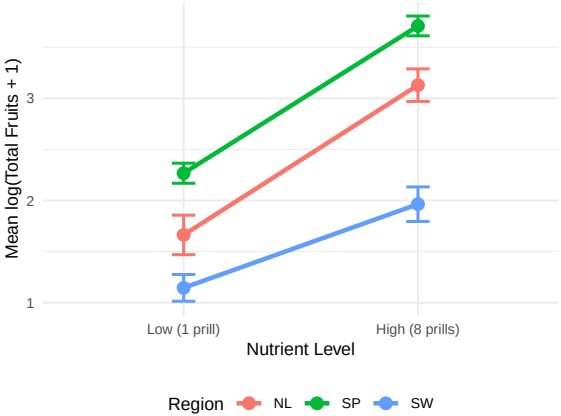
Contrast	Est.	SE	p
clipped - unclipped	-0.324	0.097	0.001

Status within Region

Reg	Contrast	Est.	SE	p
NL	Normal - Petri.Plate	-0.180	0.294	0.813
NL	Normal - Transplant	0.402	0.294	0.358
NL	Petri.Plate - Transplant	0.582	0.360	0.239
SP	Normal - Petri.Plate	0.417	0.204	0.103
SP	Normal - Transplant	0.370	0.182	0.106
SP	Petri.Plate - Transplant	-0.047	0.244	0.979
SW	Normal - Petri.Plate	0.636	0.287	0.069
SW	Normal - Transplant	-0.113	0.188	0.821
SW	Petri.Plate - Transplant	-0.749	0.312	0.044

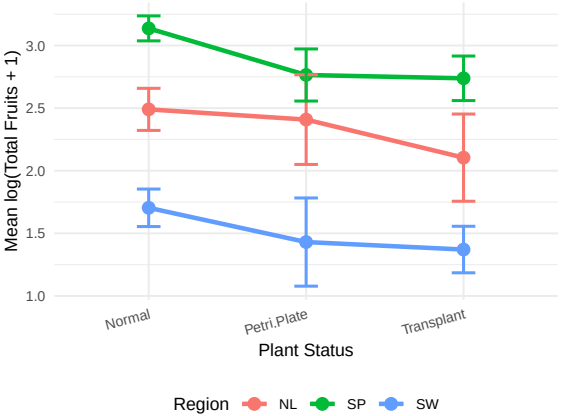
11. Interaction Plots

Region × Nutrients



- All regions benefit from high nutrients
- SP: steepest increase; SW: smallest response

Region × Status



- SP: Normal plants outperform Petri.Plate
- SW: Petri.Plate lowest; NL: less differentiation

12. Robustness: Negative Binomial GLMM

- Total Fruits is a zero-heavy count — fit a **negative binomial GLMM** on the original scale
- Same fixed / random effects as the reduced LMM
- **Result:** conclusions for key fixed effects align with the LMM; LMM retained for interpretability

Table 10: Type III tests: NB GLMM

Effect	Chisq	Df	p-value
(Intercept)	356.48	1	<0.0001
reg	7.59	2	0.0225
nutrient	107.01	1	<0.0001
status	2.23	2	0.3279
amd	8.12	1	0.0044
rack	80.31	1	<0.0001
reg:nutrient	0.63	2	0.7287
reg:status	7.08	4	0.1318

13. Conclusions

Summary

- **Nutrients, AMD, Rack, Status** are significant; high nutrients and absence of AMD $\rightarrow$ more fruits
- **Region alone** is not significant as a main effect but matters through **Region** $\times$ **Nutrients** and **Region** $\times$ **Status** interactions

Answers to research questions

1. *AMD response across region / nutrients / status:* No significant interactions between AMD and those factors — AMD's effect is consistent across them.
2. *Other interactions:* **Region** $\times$ **Nutrients** and **Region** $\times$ **Status** are significant — nutrient and transplant-source effects vary by region.
3. *Hierarchical variance:* $\approx$ 16.3% of variance attributed to populations; seed-family variance not significant.

14. Limitations, Future Work, & References

Limitations & future work

- Mild heteroscedasticity across Region / Status (variance ratios < 1.5)
- Zero-heavy count response $\rightarrow$ Gaussian LMM on $\log(y+1)$ is approximate; NB GLMM corroborates conclusions but should be the primary model in future work

References

- Banta, J. A., Stevens, M. H. H., & Pigliucci, M. (2010). *A comprehensive test of the "limiting resources" framework applied to plant tolerance to apical meristem damage.* Oikos, 119(2), 359–369.
- Dean, A., Voss, D., & Draguljić, D. (2017). *Design and Analysis of Experiments* (2nd ed.). Springer.
- STAT 642 Course Handouts 6 (Random Effects), 9 (Mixed Models and Variance Components), 10 (Model Construction and Inference).

Backup Slides

Reference material for Q&A

B1. Full Model: Type III ANOVA

Before reduction, the full LMM includes all treatment interactions up to 3-way and all nuisance interactions with Status.

Table 11: Type III F-tests for the full model

Effect	F	df1	df2	p-value
reg	2.776	2	6.43	0.135
nutrient	113.714	1	592.87	<0.001
amd	7.461	1	591.47	0.006
status	2.842	2	597.16	0.059
rack	57.630	1	589.59	<0.001
reg:nutrient	4.791	2	591.06	0.009
reg:amd	0.218	2	588.63	0.804
nutrient:amd	3.703	1	589.09	0.055
reg:status	2.335	4	596.89	0.054
nutrient:status	0.430	2	592.69	0.651
amd:status	0.485	2	592.11	0.616
reg:nutrient:amd	1.220	2	588.69	0.296

B2. Model Reduction via AIC / BIC

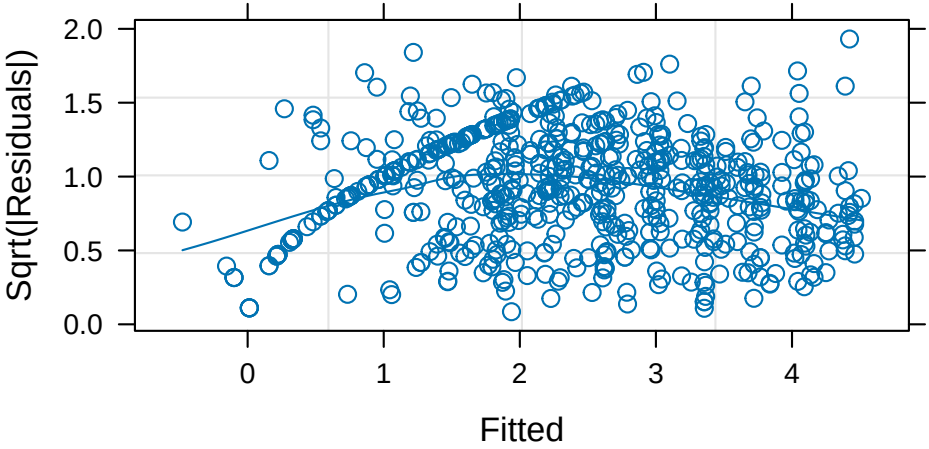
- Enumerate all subsets of 7 candidate interactions
- **Hierarchy enforced**: if 3-way is in, all three 2-way interactions must also be in
- Main effects always retained; Rack additive
- 72 models (9 treatment $\times$ 8 nuisance); fit with ML
- AIC-preferred model adds **Nutrients $\times$ AMD** vs. the LRT-reduced model; not significant by LRT
- Retain LRT-reduced model for interpretability

Table 12: Top 10 of 72 models, ranked by AIC

Model	AIC	BIC	dAIC	dBIC
Reg x Nut + Nut x AMD + Reg x Stat	2058.91	2138.79	0.00	24.19
Reg x Nut + Reg x Stat	2059.35	2134.79	0.44	20.19
Reg x Nut	2060.92	2118.61	2.01	4.01
Reg x Nut + Nut x AMD	2060.94	2123.07	2.04	8.47
Reg x Nut + Nut x AMD + Reg x Stat + AMD x Stat	2061.84	2150.60	2.93	36.00
Reg x Nut + Nut x AMD + Reg x Stat + Nut x Stat	2062.03	2150.79	3.12	36.19
Reg x Nut + Reg x Stat + AMD x Stat	2062.26	2146.58	3.35	31.98
Reg x Nut + Reg x Stat + Nut x Stat	2062.44	2146.76	3.54	32.16
Reg x Nut + Reg x AMD + Nut x AMD + Reg x Stat	2062.49	2151.25	3.58	36.65
Reg x Nut + Reg x AMD + Reg x Stat	2062.93	2147.25	4.02	32.65

B3. Scale-Location Plot

- Slight increase in variance at moderate fitted values
- No severe heteroscedasticity pattern

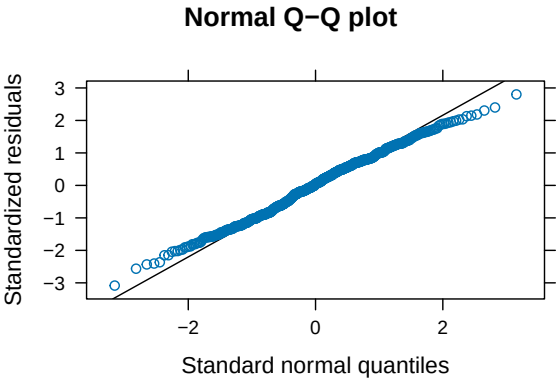


B4. Normality Detail: Shapiro-Wilk

```
    Shapiro-Wilk normality test

data:  resids
W = 0.99466, p-value = 0.02797
```

- With $n = 625$, Shapiro-Wilk is sensitive to tiny departures from normality.
- Q-Q plot is the primary diagnostic; residuals approximately normal.
- The mild non-normality is attributed to zero-inflation and is addressed indirectly by the NB GLMM robustness check.



B5. Variance Diagnostics Detail

Levene (Brown-Forsythe) test on residuals

Factor	p_value
Region	0.0059
Nutrients	0.7100
Status	0.0245
AMD	0.3870

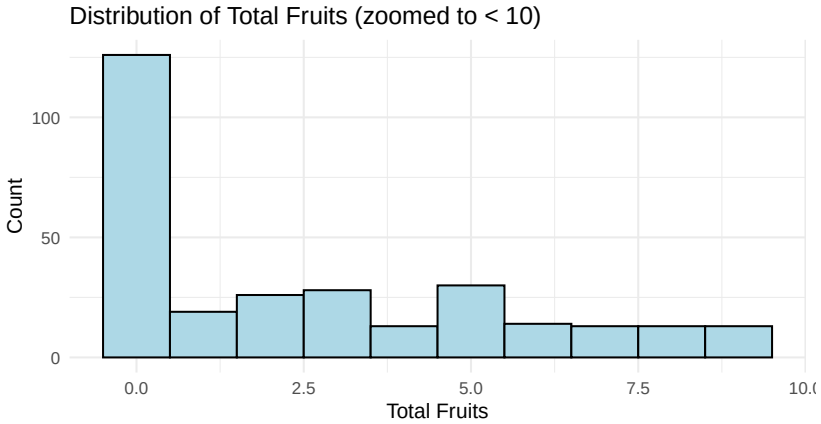
Variance ratios

Group	Variance_Ratio
Region	1.38
Status	1.43

Interpretation

- Levene detects some heteroscedasticity across Region and Status
- Variance ratios $< 1.5 \rightarrow$ not severe
- Large n makes the test sensitive to small differences
- Log transformation addresses skewness and zeros
- NB GLMM confirms robustness of conclusions

B6. Response Distribution (Zoomed)



- Heavy mass at zero motivates the $\log(y+1)$ transformation and the NB GLMM robustness check.

B7. Sampling Hierarchy Detail

Hierarchy

- 3 regions: NL, SP, SW
- 9 populations nested in Region (4 SP, 2 NL, 3 SW)
- 24 maternal seed families nested in Population

Procedure

- Seeds germinated in petri dishes, then potted
- Failed germinations replaced with same-family seedlings
- Clipping (AMD) applied at bolting stage

Random-effect specification

- `(1 | popu)` — Population random intercept; levels uniquely labeled so nesting in Region is implicit
- `(1 | gen)` — Seed family random intercept; levels uniquely labeled so nesting in Population is implicit

Scale

- $n = 625$ plants
- $\approx$ 6.5 plants per seed-family $\times$ treatment cell